A LOCAL PROPERTY WITH THE CANCELLATION PROPERTY IS INHERITED BY THE SECOND OF TWO COMPOSABLE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the author's own indexing slip, writing one of the affine formal algebraic spaces as Y_{jk} where the preceding sentence introduced Y_{kj}; the text is reproduced verbatim and no correction was inserted.

Editorial note. The counted spans carry 6 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The equivalent conditions defining the property, and their locality. Same chapter, Lemma lemma-property-defines-property-morphisms, invoked at proof lines 5411-5413. Allows the property to be tested on the members of a covering.
- The definition of a formal algebraic space through a presentation. Same chapter, Definition definition-formal-algebraic-space, invoked at proof lines 5399-5407. Supplies the three coverings used in the reduction.

Source reference. Original lemma 0ANG: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 5104–5104. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is a *formal algebraic space* if there exist a family of maps $\{X_i \to X\}_{i\in I}$ of sheaves such that

(1) X_i is an affine formal algebraic space,
(2) $X_i \to X$ is representable by algebraic spaces and étale,
(3) $\coprod X_i \to X$ is surjective as a map of sheaves

and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of formal algebraic spaces* over S is a map of sheaves.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let P be a local property of morphisms of $WAdm^{count}$ which has the cancellation property. Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of locally countably indexed formal algebraic spaces over S. If $g \circ f$ and g satisfies the equivalent conditions of Lemma 0ANG then so does $f : X \to Y$.*

Proof. Choose a covering $\{Z_k \to Z\}$ as in Definition 0.1. For each k choose a covering $\{Y_{kj} \to Z_k \times_Z Y\}$ as in Definition 0.1. For each k and j choose a covering $\{X_{kji} \to Y_{kj} \times_Y X\}$ as in Definition 0.1. Let $X_{kji} \to Y_{jk}$ and $Y_{jk} \to Z_k$ correspond to arrows $B_{kj} \to A_{kji}$ and $C_k \to B_{kj}$ of WAdm^{count}. If $g \circ f$ and g satisfies the equivalent conditions of Lemma 0ANG then $C_k \to B_{kj}$ and $C_k \to A_{kji}$ satisfy P. Hence $B_{kj} \to A_{kji}$ does too and we conclude. □